A Detailed Technical Report on the Speech Dynamics and Classification Model

Spectrogram Representation, Physics-Inspired Dynamics, Vision Transformer Classification, Domain-Adversarial Learning, and Training Strategies

Rishant Mallick

August 15, 2026

Abstract

This report presents a detailed conceptual description of the speech model discussed in this work. The model treats speech as a structured time-frequency acoustic signal and, in the physics-inspired component, as a dynamical system driven by an excitation. The speech spectrogram contains energy distributed over multiple frequency regions. Because different parts of the vocal tract have different acoustic effects, the overall vocal-tract configuration determines how acoustic energy is distributed across frequency.

A convolutional stem can extract local time-frequency patterns and reduce the input resolution before a Vision Transformer (ViT) models long-range relationships between time-frequency patches. A CLS token provides a global representation for classification. In the dynamical branch, a shared driving force $F(t)$ is combined with frequency-dependent damping $\gamma(f,t)$ and oscillatory parameters $\omega(f,t)$. Smoothness and gradient-difference losses can regularize the driving force so that it does not become unnecessarily irregular.

For domain-adversarial learning, a Gradient Reversal Layer (GRL) connects the learned representation to a domain classifier. The domain classifier attempts to identify the data domain, while the reversed gradient encourages the encoder to learn features that are useful for the main task but less dependent on domain-specific characteristics. Stochastic depth, GELU activation, stride-based downsampling, and cosine learning-rate scheduling are also discussed. The report emphasizes the distinction between acoustic frequency components and physical anatomical regions and clarifies that GRL is not a feature-relevance or final classification mechanism.

Contents

1 Introduction and Motivation **3**

2 Speech as a Time-Frequency Signal **3**
2.1 Spectrogram 3
2.2 Frequency Components and the Vocal Tract 3

3 Physics-Inspired Dynamical Interpretation **4**

4 Why F, γ, and ω Have Different Outputs **4**
4.1 Driving Force 4
4.2 Frequency-Dependent Damping 5
4.3 Frequency-Dependent Oscillatory Parameters 5

5 Physical and Acoustic Intuition **5**

6 Smoothness and Gradient-Difference Losses **6**
6.1 Smoothness Loss 6
6.2 Gradient-Difference Loss 6

7 Overall Feature-Extraction Pipeline **6**

8 Convolutional Stem **7**

9 Stride and Downsampling **7**

10 GELU Activation **7**

11 Vision Transformer as the Classifier **8**
11.1 Why Use ViT? 8

12 CLS Token and Final Classification **8**

13 Stochastic Depth **9**

14 Gradient Reversal Layer **9**
14.1 Purpose 9
14.2 What Is a Domain? 10
14.3 Forward Pass 10
14.4 Backward Pass 10

15 How GRL Training Works **10**

16 What GRL Does Not Do **11**

17 Final Prediction Versus Domain Prediction **11**
17.1 Main Prediction 12
17.2 Domain Prediction 12

18 Feature Extraction with GRL **12**

19 Cosine Learning-Rate Schedule **12**

20 Reconstruction and Prediction **13**

21 Why the Model Is Not Necessarily Causal **13**

22 Causal Positional Encoding **14**

23 Sine-Cosine Positional Encoding **14**

24 Training Losses **14**

25 Conceptual End-to-End Architecture **15**

26 Summary Table **15**

27 Key Conceptual Takeaways **15**

28 Compact Description for a Presentation **17**

29 Conclusion **17**

1 Introduction and Motivation

The central objective of the model is to learn meaningful representations of speech while capturing both its acoustic structure and temporal dynamics. Speech is not a static signal. It evolves over time as the vocal folds generate excitation and the vocal tract filters and shapes that excitation.

A useful high-level representation is

$$\boxed{\text{Speech} \rightarrow \text{Spectrogram} \rightarrow \text{Feature Encoder} \rightarrow \text{Learned Representation} \rightarrow \text{Prediction}}$$

The architecture can additionally include a dynamical reconstruction branch and a domain-adversarial branch.

The motivation for reconstruction or dynamical modeling is particularly important when labeled data are limited. A model can first learn useful speech representations from reconstruction or other self-supervised objectives and subsequently use those representations for supervised classification. Thus, reconstruction and prediction are not contradictory: reconstruction can help the model learn speech dynamics, while prediction uses the learned representation for the final task.

2 Speech as a Time-Frequency Signal

2.1 Spectrogram

A spectrogram represents acoustic information simultaneously along two dimensions:

- time,
- frequency.

Let

$$X(f, t)$$

denote the spectrogram magnitude or energy at frequency f and time t.

If the spectrogram has 50 frequency bins, a single time frame can be represented as

$$X_t = [X(f_1, t), X(f_2, t), \ldots, X(f_{50}, t)].$$

Thus, the 50 values are sampled values from 50 different frequency regions.

These values are not 50 measurements of 50 physical vocal-tract parts. Instead, they describe the acoustic energy present in different portions of the frequency axis at a particular instant.

2.2 Frequency Components and the Vocal Tract

A suitable description of the acoustic interpretation is:

> Different parts of the vocal tract have different acoustic effects, and the overall shape and configuration of the vocal tract determine how acoustic energy is distributed across different frequency regions.

The tongue, lips, jaw, pharyngeal structures, and other articulators modify the geometry of the vocal tract. The resulting configuration affects acoustic resonances and therefore changes the spectral distribution.

The relationship can be summarized as

Articulator movement $\rightarrow$ Vocal-tract configuration $\rightarrow$ Acoustic response $\rightarrow$ Frequency distribution.

It would therefore be inaccurate to claim that one particular frequency bin corresponds directly to one anatomical part.

Instead,

$$\boxed{\text{one vocal-tract configuration} \rightarrow \text{many frequency components}}$$

and one articulator can influence several frequency components simultaneously.

3 Physics-Inspired Dynamical Interpretation

The model uses the intuition of a damped dynamical system. A generic second-order form can be written as

$$\frac{d^2 X_f}{dt^2} + 2\gamma_f \frac{dX_f}{dt} + \omega_f^2 X_f = F. \tag{1}$$

This equation should be interpreted as a learned dynamical abstraction of the acoustic signal, rather than as a literal statement that every spectrogram bin is an independent physical oscillator.

The terms have the following conceptual roles:

- F is the driving force or excitation.
- γ controls damping-related behavior.
- ω controls oscillatory or natural-frequency-related behavior.
- X_f represents the response associated with a frequency component.

The vocal folds can be viewed as providing an excitation that drives the acoustic system, while the vocal tract shapes the resulting response.

4 Why F, γ, and ω Have Different Outputs

One important architectural choice is that the F-head can produce one value while the γ-head and ω-head can produce 50 values.

The reason is an architectural modeling assumption: F is treated as a shared excitation, whereas γ and ω are allowed to vary across frequency bins.

4.1 Driving Force

The force can be represented as

$$F(t).$$

It is treated as a common driving input to the dynamical system:

$$\boxed{F(t) = \text{shared driving/excitation signal.}}$$

4.2 Frequency-Dependent Damping

The damping vector can be written as

$$\boldsymbol{\gamma}(t) = [\gamma_1(t), \gamma_2(t), \ldots, \gamma_{50}(t)].$$

Each element corresponds to one frequency bin:

$$\gamma_i(t) = \gamma(f_i, t).$$

This allows different frequency components to have different rates of temporal decay.

4.3 Frequency-Dependent Oscillatory Parameters

Similarly,

$$\boldsymbol{\omega}(t) = [\omega_1(t), \omega_2(t), \ldots, \omega_{50}(t)]$$

where

$$\omega_i(t) = \omega(f_i, t).$$

This allows different frequency regions to exhibit different learned oscillatory dynamics.
Therefore,

$$\boxed{F(t) \to \text{shared excitation}}$$

while

$$\boxed{\gamma(f,t), \omega(f,t) \to \text{frequency-dependent dynamics.}}$$

The 50 values of γ and ω are associated with the 50 acoustic frequency bins, not with 50 anatomical regions.

5 Physical and Acoustic Intuition

At a given instant, speech contains energy distributed over multiple frequency components:

$$X(t) = [X(f_1, t), \ldots, X(f_{50}, t)].$$

As time progresses, the energy in these components can evolve differently. For example, one frequency region might decay quickly while another persists for longer.

Conceptually,

$$X(f, t_1) > X(f, t_2)$$

indicates a decrease in the magnitude of that component between t_1 and t_2.

The damping parameter can therefore capture the rate of temporal decay:

$$\gamma(f, t) \to \text{damping behavior of frequency component } f.$$

The important point is that this does not mean that a physical part of the vocal tract moves from one frequency to another. Rather, the acoustic energy is distributed across frequency components, and those components can have different temporal dynamics.

6 Smoothness and Gradient-Difference Losses

A learned driving force can become unnecessarily irregular if it is optimized only for reconstruction or prediction. Regularization can encourage a physically and temporally reasonable trajectory.

6.1 Smoothness Loss

A simple smoothness loss is

$$L_{\text{smooth}} = \sum_t |F_{t+1} - F_t|^2 . \tag{2}$$

This penalizes large frame-to-frame changes in F.
Therefore,

$$\boxed{L_{\text{smooth}} \rightarrow \text{penalizes rapid changes in } F.}$$

It does not force speech to be perfectly smooth. Instead, it discourages unnecessary jaggedness in the learned driving signal.

6.2 Gradient-Difference Loss

A gradient-difference loss can be written as

$$L_{\text{grad\_diff}} = \sum_t [(F_{t+1} - F_t) - (F_t - F_{t-1})]^2 . \tag{3}$$

This is equivalent to

$$L_{\text{grad\_diff}} = \sum_t [F_{t+1} - 2F_t + F_{t-1}]^2 . \tag{4}$$

It therefore penalizes large second differences.
The distinction is

$$\boxed{L_{\text{smooth}} : \text{controls first-order temporal variation}}$$

whereas

$$\boxed{L_{\text{grad\_diff}} : \text{controls changes in the temporal slope.}}$$

Together they encourage a well-behaved temporal trajectory.

7 Overall Feature-Extraction Pipeline

A conceptual version of the model is

$$\boxed{\text{Speech} \rightarrow \text{Spectrogram} \rightarrow \text{ConvStem} \rightarrow \text{GELU} \rightarrow \text{Stride Downsampling} \rightarrow \text{Patch/Token Representation} \rightarrow \text{V}}$$

The resulting representation can feed multiple heads:

$$z = h_{\text{CLS}}.$$

Then

$$z \rightarrow \text{HC/PD Classification Head},$$

while a separate branch can use

$$z \to \text{GRL} \to \text{Domain Classifier}.$$

The dynamical heads can also be attached to an appropriate learned representation:

$$z \to \begin{cases} F\text{-head} \\ \gamma\text{-head} \\ \omega\text{-head}. \end{cases}$$

8 Convolutional Stem

Before applying a Transformer, a convolutional stem can process the spectrogram:

$$\text{Spectrogram} \to \text{Convolution} \to \text{GELU} \to \text{Convolution} \to \text{Downsampling}.$$

The convolutional stem has two major roles:

1. extract local time-frequency patterns;
2. reduce the input resolution before self-attention.

Convolutions are naturally suited to local structures. For spectrograms, these can include localized patterns across time and frequency.

9 Stride and Downsampling

A convolution with stride greater than one can reduce spatial or time-frequency resolution.

For example,

$$224 \times 224 \to 112 \times 112$$

with stride 2.

A second stride-2 operation can produce

$$112 \times 112 \to 56 \times 56.$$

The exact dimensions depend on kernel size, padding, and stride, but the main idea is that downsampling reduces the number of tokens.

This is important because self-attention becomes more expensive as the number of tokens increases.

Thus,

$$\boxed{\text{ConvStem} + \text{stride} \to \text{local features} + \text{fewer tokens}.}$$

10 GELU Activation

GELU is the Gaussian Error Linear Unit:

$$\text{GELU}(x) = x\Phi(x), \tag{5}$$

where $\Phi(x)$ is the standard Gaussian cumulative distribution function.

Unlike ReLU,

$$\mathrm{ReLU}(x) = \max(0, x),$$

GELU behaves smoothly and softly weights the input.

Its role is to provide nonlinear transformations so that the network can learn complex feature mappings. It does not by itself perform frequency analysis.

11 Vision Transformer as the Classifier

Although a Vision Transformer was originally developed for images, a spectrogram is also a two-dimensional representation. Its two axes are time and frequency rather than image height and width.

The basic pipeline is

$$\text{Spectrogram} \rightarrow \text{Patches} \rightarrow \text{Patch Embeddings} \rightarrow \text{Transformer}.$$

Each patch is converted into a vector. The resulting sequence is

$$[P_1, P_2, \ldots, P_N].$$

The Transformer then uses self-attention to model relationships between these patches.

11.1 Why Use ViT?

The convolutional stem is effective at local feature extraction, while the Transformer can model long-range relationships.

Therefore,

$$\boxed{\text{ConvStem} \rightarrow \text{local patterns}}$$

and

$$\boxed{\text{ViT} \rightarrow \text{global relationships.}}$$

For speech, this means the model can learn relationships between distant time-frequency regions rather than relying only on local neighborhoods.

12 CLS Token and Final Classification

A special learnable token called the CLS token can be prepended to the sequence:

$$[CLS, P_1, P_2, \ldots, P_N].$$

Self-attention allows the CLS token to interact with all other tokens.

After the Transformer,

$$h_{\mathrm{CLS}}$$

can be used as a global representation of the input.

The classification pipeline is

$$\text{Spectrogram} \rightarrow \text{Patch Embeddings} \rightarrow [CLS, P_1, \ldots, P_N] \rightarrow \text{ViT} \rightarrow h_{\mathrm{CLS}} \rightarrow \text{Classifier}.$$

For binary classification,

$$h_{\text{CLS}} \to \text{Linear Layer} \to [P(HC), P(PD)].$$

If, for example,

$$P(HC) = 0.30, \qquad P(PD) = 0.70,$$

the final prediction is PD.

The final HC/PD prediction therefore comes from the main classification head, not from the GRL branch.

13 Stochastic Depth

Stochastic depth is a regularization technique in which complete residual blocks are randomly skipped during training.

Suppose a network contains

$$B_1, B_2, B_3, B_4.$$

One training iteration might use

$$B_1 \to B_2 \to B_3 \to B_4,$$

while another might skip B_3:

$$B_1 \to B_2 \to B_4.$$

The purpose is to prevent the network from becoming excessively dependent on particular blocks and to improve generalization.

It differs from ordinary dropout. Dropout typically masks individual activations, whereas stochastic depth drops an entire residual path or computational block.

14 Gradient Reversal Layer

14.1 Purpose

The Gradient Reversal Layer is used for domain-adversarial learning.

Suppose the ViT encoder produces

$$z.$$

The representation is sent to two branches:

$$z \to \text{HC/PD Classifier}$$

and

$$z \to \text{GRL} \to \text{Domain Classifier}.$$

The first branch is responsible for the final disease prediction.

The second branch is used to make the learned representation less dependent on domain-specific information.

14.2 What Is a Domain?

A domain is a source or environment associated with systematic differences in the data.

For speech, examples can include

- different datasets,
- different microphones,
- different recording environments,
- different recording sessions,
- other systematic acquisition differences.

For example, suppose Dataset A was recorded using Microphone A and Dataset B using Microphone B. The model could accidentally learn microphone-specific patterns instead of disease-relevant speech characteristics.

An undesirable shortcut is

$$\text{microphone characteristics} \to \text{HC/PD prediction.}$$

The desired behavior is

$$\text{speech characteristics} \to \text{HC/PD prediction.}$$

14.3 Forward Pass

During the forward pass, GRL behaves approximately as the identity:

$$\mathrm{GRL}(z) = z. \tag{6}$$

Therefore, GRL does not itself extract features.

14.4 Backward Pass

During backpropagation,

$$\frac{\partial\,\mathrm{GRL}}{\partial z} = -\lambda I. \tag{7}$$

If the incoming gradient is g, the encoder receives

$$-\lambda g.$$

This is the defining behavior of gradient reversal.

15 How GRL Training Works

Let the main classification loss be

$$L_{\text{class}}.$$

The main classifier tries to minimize it:

$$L_{\text{class}} \downarrow .$$

The domain classifier predicts the domain and minimizes

$$L_{\text{domain}}.$$

However, the GRL reverses the domain gradient before it reaches the feature extractor. Therefore, the two parts have competing objectives:

$$\boxed{\text{Domain classifier: predict the domain accurately}}$$

while

$$\boxed{\text{Feature extractor: make domain prediction difficult.}}$$

At the same time,

$$\boxed{\text{Feature extractor: preserve information useful for HC/PD classification.}}$$

A conceptual minimax formulation is

$$\min_{\theta_f,\theta_y} \max_{\theta_d} \left[L_{\text{class}} - \lambda L_{\text{domain}}\right], \tag{8}$$

where θ_f represents feature-extractor parameters, θ_y the main classifier parameters, and θ_d the domain-classifier parameters. Equivalent implementations may express the optimization using a gradient-reversal operation rather than explicitly writing a minimax optimizer.

16 What GRL Does Not Do

GRL should not be interpreted as a mechanism that assigns a relevance probability to every feature.

Suppose

$$z = [z_1, z_2, \dots, z_d].$$

The domain classifier may output

$$P(D_1), P(D_2), \dots$$

which are probabilities of domains.

These are **domain probabilities**, not probabilities of feature importance.

GRL only modifies the gradient:

$$g \to -\lambda g.$$

Therefore:

$$\boxed{\text{Domain classifier} \to \text{domain probabilities}}$$

whereas

$$\boxed{\text{GRL} \to \text{gradient reversal.}}$$

If feature relevance is required for interpretation, other methods such as attention analysis, gradients, integrated gradients, or SHAP can be used.

17 Final Prediction Versus Domain Prediction

There are two different prediction tasks.

17.1 Main Prediction

The main branch produces

$$[P(HC), P(PD)].$$

This determines the final HC/PD prediction.

17.2 Domain Prediction

The domain branch produces something such as

$$[P(D_1), P(D_2)].$$

This prediction is used to generate the adversarial learning signal.

Thus,

$$\boxed{\text{GRL branch does not determine the final HC/PD class.}}$$

Instead, it helps shape the representation used by the main classifier.

18 Feature Extraction with GRL

The GRL does not extract the features itself. The feature extractor is composed of the convolutional and Transformer components.

Training repeatedly performs

$$\text{input} \to \text{encoder} \to z \to \text{predictions} \to \text{loss} \to \text{backpropagation}.$$

The HC/PD loss encourages features that make the disease classification easier.

The domain loss, after passing through GRL, encourages the encoder to remove or suppress information that makes the domain easy to identify.

Ideally, the representation becomes

$$z = \underbrace{\text{task-relevant speech information}}_{\text{retain}} + \underbrace{\text{domain-specific information}}_{\text{reduce}}.$$

The objective is therefore not to remove all information. It is to reduce nuisance information while preserving information required for the main task.

19 Cosine Learning-Rate Schedule

A cosine learning-rate schedule changes the learning rate smoothly over training.

A common formulation is

$$LR_t = LR_{\min} + \frac{1}{2}(LR_{\max} - LR_{\min}) \left[1 + \cos\left(\frac{\pi t}{T}\right)\right], \tag{9}$$

where

- t is the current epoch or training step,
- T is the total number of epochs or steps,
- $LR_{\max}$ is the initial learning rate,

- $LR_{\min}$ is the minimum learning rate.

For example, with

$$LR_{\max} = 10^{-3}, \qquad LR_{\min} = 0,$$

the approximate values are

Epoch	Learning Rate
0	0.00100
25	0.00085
50	0.00050
75	0.00015
100	0.00000

The learning rate therefore starts relatively large and becomes increasingly small.
This allows the model to make larger updates early in training and smaller refinements later.
Importantly,

$$\boxed{\text{cosine scheduling controls the learning rate, not the loss itself.}}$$

20 Reconstruction and Prediction

The model can use reconstruction as a representation-learning objective before or alongside the supervised prediction task.

The conceptual idea is

$$\text{speech} \rightarrow \text{encoder} \rightarrow \text{latent representation} \rightarrow \text{reconstruction}.$$

The reconstruction objective forces the representation to retain useful information about the speech signal.

When labeled data are scarce, this can be valuable because unlabeled speech can contribute to learning the underlying dynamics.

After learning the representation, a classifier can use it:

$$\text{learned representation} \rightarrow \text{HC/PD classifier}.$$

Therefore, reconstruction can be viewed as a way of learning speech dynamics, while classification is the final prediction task.

21 Why the Model Is Not Necessarily Causal

A causal model can only use information from the past when producing the current output:

$$x_t \leftarrow x_1, \ldots, x_t.$$

A non-causal model can use both past and future context:

$$x_t \leftarrow x_{t-k}, \ldots, x_t, \ldots, x_{t+k}.$$

For speech reconstruction or analysis of speech dynamics, both past and future context can be useful because the acoustic realization around a time point is influenced by transitions before and after that point.

This is analogous to the motivation for bidirectional sequence models such as BiLSTMs.

If the objective is truly real-time prediction with strict latency constraints, future frames cannot be used beyond the allowed latency. Therefore, whether the model should be causal depends on the operational definition of "real time."

For offline reconstruction or representation learning, a non-causal Transformer can use both temporal directions.

22 Causal Positional Encoding

There is no separate mechanism that must be called "causal positional encoding."

Positional encoding provides information about token position:

$$P_i.$$

Causality is instead imposed through the attention pattern, typically by using an attention mask that prevents a token at time t from attending to future positions.

Thus,

$$\boxed{\text{positional encoding} \neq \text{causal masking.}}$$

A Transformer can use positional information without being causal.

For a non-causal speech reconstruction model, positional encoding can be used normally without a causal attention mask.

23 Sine-Cosine Positional Encoding

A standard sinusoidal positional encoding can be written as

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right), \tag{10}$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right). \tag{11}$$

Here:

- pos is the token position,
- i is the index of the embedding dimension pair,
- d_{model} is the embedding dimension.

The variable i does **not** represent time. It indexes different dimensions of the positional encoding.

Different dimensions use different frequencies, allowing the Transformer to infer relative positions from combinations of sine and cosine signals.

24 Training Losses

A general multi-objective model can be described by

$$L_{\text{total}} = L_{\text{task}} + \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{smooth}} L_{\text{smooth}} + \lambda_{\text{grad}} L_{\text{grad\_diff}} + \lambda_{\text{adv}} L_{\text{domain}}, \tag{12}$$

with the important qualification that the domain term is implemented adversarially through the GRL rather than simply minimized by the feature extractor.

The exact coefficients and loss components depend on the final implementation.

25 Conceptual End-to-End Architecture

The complete conceptual architecture can be summarized as

$$\boxed{\begin{array}{c}
\text{Speech waveform} \\
\downarrow \\
\text{Spectrogram} \\
\downarrow \\
\text{ConvStem} \\
\downarrow \\
\text{GELU + Stride Downsampling} \\
\downarrow \\
\text{Patch / Token Embeddings} \\
\downarrow CLS, P_1, \ldots, P_N > \text{CLS,P}_1, \ldots, P_N \\
\\
\downarrow \\
\text{Vision Transformer} \\
\downarrow \\
h_{\text{CLS}}
\end{array}}$$

From the learned representation, the heads can be represented as

$$h_{\text{CLS}} \to \begin{cases} \text{HC/PD classifier}, \\ \text{GRL} \to \text{domain classifier}, \\ F\text{-head}, \\ \gamma\text{-head}, \\ \omega\text{-head}. \end{cases}$$

The exact connection of the dynamical heads can depend on the implementation, but the conceptual purpose remains the same.

26 Summary Table

27 Key Conceptual Takeaways

1. The 50 spectrogram values are 50 acoustic frequency bins, not 50 anatomical parts of the vocal tract.

2. Different parts of the vocal tract have different acoustic effects, and the overall vocal-tract configuration determines how acoustic energy is distributed across frequency.

3. The vocal folds can be interpreted as providing excitation, while the vocal tract shapes the acoustic response.

4. F is modeled as a shared driving force in the discussed architecture, whereas γ and ω are frequency-bin dependent.

5. γ describes damping-related behavior, while ω describes oscillatory or natural-frequency-related behavior.

6. Smoothness loss penalizes rapid changes in F, while gradient-difference loss penalizes rapid changes in the slope of F.

7. The convolutional stem extracts local patterns and performs downsampling.

Component	Function	Purpose
Spectrogram	Represents acoustic energy over time and frequency	Provides structured time-frequency input
ConvStem	Extracts local time-frequency patterns	Captures local acoustic structure
Stride	Reduces resolution	Reduces token count and computation
GELU	Smooth nonlinear activation	Enables complex feature transformations
ViT	Self-attention over patches	Captures long-range relationships
CLS	Global Transformer representation	Provides a compact representation for classification
F-head	Predicts shared driving force	Represents common excitation
γ-head	Predicts 50 frequency-dependent damping values	Allows different spectral components to decay differently
ω-head	Predicts 50 frequency-dependent oscillatory values	Allows different spectral components to have different dynamics
L_{smooth}	Penalizes rapid changes in F	Encourages temporal smoothness
$L_{\text{grad\_diff}}$	Penalizes rapid changes in the slope of F	Encourages smoother dynamics
GRL	Reverses domain gradients	Encourages domain-invariant features
Domain classifier	Predicts domain	Provides adversarial training signal
Stochastic depth	Randomly skips blocks during training	Regularizes deep networks
Cosine scheduler	Gradually reduces learning rate	Supports stable optimization and late-stage refinement

Table 1: Summary of the major model components discussed.

8. The ViT models global relationships between time-frequency patches.
9. The CLS token provides a global representation that can be passed to the HC/PD classification head.
10. GRL does not classify HC/PD and does not directly provide feature relevance. It reverses the domain gradient.
11. The domain classifier predicts the domain, while GRL causes the feature extractor to learn representations in which the domain is harder to detect.
12. The final HC/PD prediction comes from the main disease classification head.
13. Stochastic depth is a regularization method that drops complete computational paths rather than individual activations.
14. A cosine learning-rate scheduler changes the learning rate over training; it does not directly change the definition of the loss.

15. Reconstruction can be used to learn speech dynamics and representations when labeled data are limited, after which the learned representation can support supervised prediction.

28 Compact Description for a Presentation

The model can be described concisely as follows:

> The model represents speech using a time-frequency spectrogram. A convolutional stem first extracts local acoustic patterns and performs stride-based downsampling. The resulting time-frequency patches are processed by a Vision Transformer, whose self-attention captures long-range relationships across the spectrogram. A CLS token summarizes the learned representation and is used for HC/PD classification.
>
> In the physics-inspired dynamical branch, the vocal-fold excitation is represented by a shared driving force F, while damping γ and oscillatory parameter ω are predicted separately for the 50 frequency bins because different acoustic frequency components can exhibit different temporal dynamics. Smoothness and gradient-difference losses regularize the learned driving force.
>
> A Gradient Reversal Layer can be attached to a domain classifier to encourage the encoder to learn domain-invariant representations. The domain classifier tries to identify the recording domain, while the reversed gradient encourages the encoder to make that domain harder to detect. The GRL does not produce the final HC/PD prediction; that prediction comes from the main classification head. Stochastic depth provides regularization, while a cosine learning-rate schedule gradually reduces the learning rate during optimization.

29 Conclusion

The discussed architecture combines several complementary ideas. The spectrogram provides a structured representation of speech in the time-frequency domain. The convolutional stem learns local acoustic patterns efficiently, while the ViT captures global relationships among distant time-frequency regions. The CLS representation can then support the final classification task.

The physics-inspired branch provides an interpretable dynamical viewpoint. The shared force F represents excitation, while frequency-dependent γ and ω allow the model to describe different temporal behaviors across spectral components. Smoothness-based regularization helps keep the learned dynamics well behaved.

The domain-adversarial component serves a different purpose. GRL does not extract features or assign probabilities to individual features. Instead, it changes the direction of the domain-classification gradient so that the feature extractor is encouraged to suppress domain-specific information while retaining information useful for HC/PD classification.

Together, these components provide a framework for learning speech representations that capture local acoustic structure, global time-frequency relationships, temporal dynamics, and robustness to nuisance domain variation.